

Activity-APPLICATION BUILDING USING WEKA TOOL

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1. Weather-Based Play Prediction ☁️

Build a WEKA application that predicts whether a person should play outdoors based on weather conditions.

- Dataset: Outlook, Temperature, Humidity, Windy
- Target: Play (Yes/No)
- Algorithm: **J48 Decision Tree**
- Tasks:
 1. Load the dataset into WEKA.
 2. Preprocess the data.
 3. Apply J48.
 4. Visualize the decision tree.
 5. Predict Play for a new weather condition.

2. Flower Species Classification 🌸

Build a WEKA application that identifies the species of a flower from its physical measurements.

- Dataset: Iris dataset
- Attributes: Sepal Length, Sepal Width, Petal Length, Petal Width
- Target: Iris Species
- Algorithm: **J48 or Naive Bayes**
- Tasks:
 1. Load the Iris dataset.
 2. Select the classification algorithm.
 3. Train the model.
 4. Evaluate its accuracy.
 5. Classify a new flower.

Answers:

1) Weather-Based Play Prediction

Classifier output

```
Model attributes          Incoming attributes
-----
(numeric) Weight         --> 1 (numeric) Weight
(nominal) Color          --> 2 (nominal) Color
(nominal) Size           --> 3 (nominal) Size
(numeric) Firmness       --> 4 (numeric) Firmness
(nominal) Quality        --> 5 (nominal) Quality

Time taken to build model: 0 seconds

=== Predictions on test set ===

inst#,actual,predicted,error,prediction
1,1:?,1:Good,,1
2,1:?,2:Bad,,1

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.01 seconds

=== Summary ===

Total Number of Instances          0
Ignored Class Unknown Instances    2

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                ?        ?        ?         ?        ?          ?        ?        ?        Good
                ?        ?        ?         ?        ?          ?        ?        ?        Bad
Weighted Avg.   ?        ?        ?         ?        ?          ?        ?        ?

=== Confusion Matrix ===

a b  <-- classified as
0 0 | a = Good
0 0 | b = Bad
```

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose J48 -C 0.25 -M 2

Test options

☐ Use training set

☒ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Quality

Start Stop

Result list (right-click for options)

11:25:52 - trees.J48

11:45:17 - misc.InputMappedClassifier

11:50:21 - misc.InputMappedClassifier

Classifier output

J48 pruned tree

Color = Red: Good

Color = Green: Bad

Color = Yellow: Good

Number of Leaves

Size of the tree

Time taken to build model

=== Stratified cross validation ===

=== Summary ===

Correctly Classified Instances 11

Incorrectly Classified Instances 1

Kappa statistic 0.909

Mean absolute error 0.1

Root mean squared error 0.316

Relative absolute error 63.095%

Root relative squared error 63.095%

Total Number of Instances 12

=== Detailed Accuracy By Class ===

Weighted Avg.

=== Confusion Matrix ===

a b <-- classified as

11 0 | a = Good

1 0 | b = Bad

Weka Classifier Tree Visualizer: 11:25:52 - trees.J48 (fruit_qty)

Tree View

```
graph TD
    Color((Color)) -- = Red --> Good5[Good (5.0)]
    Color -- = Green --> Bad9[Bad (9.0)]
    Color -- = Yellow --> Good6[Good (6.0)]
```

2) Flower Species Classification

Classifier output

=== Run information ===

Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2
Relation: iris
Instances: 150
Attributes: 5
SepalLength
SepalWidth
PetalLength
PetalWidth
Class
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree

PetalWidth <= 0.6: Iris-setosa (50.0)
PetalWidth > 0.6
| PetalWidth <= 1.7
| | PetalLength <= 4.9: Iris-versicolor (48.0/1.0)
| | PetalLength > 4.9
| | | PetalWidth <= 1.5: Iris-virginica (3.0)
| | | PetalWidth > 1.5: Iris-versicolor (3.0/1.0)
| PetalWidth > 1.7: Iris-virginica (46.0/1.0)

Number of Leaves : 5

Size of the tree : 9

Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	144	96	%
Incorrectly Classified Instances	6	4	%
Kappa statistic	0.94		

Mean absolute error	0.035
Root mean squared error	0.1586
Relative absolute error	7.8705 %
Root relative squared error	33.6353 %
Total Number of Instances	150

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.980	0.000	1.000	0.980	0.990	0.985	0.990	0.987	Iris-setosa
	0.940	0.030	0.940	0.940	0.940	0.910	0.952	0.880	Iris-versicolor
	0.960	0.030	0.941	0.960	0.950	0.925	0.961	0.905	Iris-virginica
Weighted Avg.	0.960	0.020	0.960	0.960	0.960	0.940	0.968	0.924	

=== Confusion Matrix ===

a	b	c	<-- classified as
49	1	0	a = Iris-setosa
0	47	3	b = Iris-versicolor
0	2	48	c = Iris-virginica

